Yumeng He

+1-801-347-7770 | u1528477@utah.edu | ymh1003
Salt Lake City, UT

Education

- **University of Utah** *August 2024 - Present*
  PhD in Computer Science Salt Lake City, UT
  - GPA: 4.00/4.00, advised by Pavel Panchekha
  - Research Interests: Program analysis, numerical computing, and formal verification.
- **University of Rochester** *August 2020 - May 2024*
  Bachelor of Science, Double Major in Computer Science and Mathematics, Minor in Business Rochester, NY
  - Grade: 3.98/4.00, graduated summa cum laude, highest distinction.

Publications

[1] **He, Y.**, & Panchekha, P. (2026). Accurate Residues for Floating-Point Debugging. *OOPSLA 2026*. [DOI] [Talk] [Artifact] [Code]

[2] **He, Y.**, Nandi, C., & Pai, S. (2025). Formalizing Linear-Motion G-code for Invariant Checking and Differential Testing of Fabrication Tools. *OOPSLA 2025*. [DOI] [Talk] [Artifact] [Code]

Professional Experience

- **ML Compiler Engineer Intern** *May 2026 - Aug 2026*
  Annapurna Labs, AWS Cupertino, CA
  - Developed compiler support for dynamic tensor shapes on AWS Trainium and Inferentia accelerators.
  - Designed and implemented an LICM pass with SSA-based safety and register-pressure checks.
  - Extended existing IR, register-allocation, and optimization passes from general-purpose to hardware shape registers.

Projects

- **Math Library Verification** *August 2026 - Present*
  - Developed a verification framework that derives sound, tight error bounds for math library implementations.
- **Efficient Numeric Debugging with Improved Accuracy [1]** *August 2024 - May 2026*
  - Improved diagnostic accuracy in an LLVM-based floating-point debugger without using high-precision arithmetic.
  - Extended a WebAssembly interpreter for shadow execution to track per-variable floating-point error.
  - Evaluated on 44 real-world workloads, showing higher accuracy on cases where baseline systems yield false results.
- **Differential Fuzzing for Bug Detection in Fabrication Tools [2]** *May 2023 - October 2025*
  - Developed a static analysis tool to detect software bugs in 3D printing pipelines using differential testing on G-code.
  - Defined formal G-code semantics for scalable geometric comparison.
  - Detected rotation-invariant violations across 58 real-world models, revealing bugs in production fabrication tools.
- **Lease Cache Design** *May 2023 - August 2023*
  - Modeled performance degradation resulting from optimal virtual lease assignments in physical cache systems.
  - Developed mathematical proofs for key properties of the model.

Honors and Awards

- **Excellence in Undergraduate Research** *May 2024*
  Department of Computer Science, University of Rochester
- **Phi Beta Kappa** *March 2023*
  University of Rochester Chapter

Academic Service

- POPL 2027 Artifact Evaluation Committee Member
- MFCS 2024 Referee

Miscellaneous

- TA for Vibe Coding, CS Dept., University of Utah *Spring 2026*
- TA for Computer Architecture, CS Dept., University of Rochester *Spring 2023*
- TA for Operations Research, Math Dept., University of Rochester *Fall 2022*